

Class 10 – Artificial Intelligence (417)

50 Important Questions with Exam-Ready Answers

PART A – EMPLOYABILITY SKILLS

Unit 1: Communication Skills-II

Q1. What is communication? Explain its importance in daily life.

Answer:

Communication is the process of exchanging information, ideas, thoughts and feelings between two or more people through speaking, writing, gestures, signs or visuals. The word comes from the Latin word "*communicare*", meaning "to share".

Importance of communication in daily life:

- **Sharing information** – helps us give and receive knowledge, news and instructions.
- **Expressing feelings and needs** – we can tell others what we want, feel or think.
- **Building relationships** – good communication creates trust and friendship in family, school and society.
- **Learning and education** – teaching and learning depend fully on communication.
- **Teamwork and problem solving** – people can discuss, plan and solve problems together.
- **Avoiding misunderstanding** – clear communication removes confusion and conflicts.

Q2. What are the different methods of communication?

Answer:

There are four main methods of communication:

1. **Verbal communication** – exchange of information using words, either spoken (talking, phone call, speech) or written (letter, email, SMS).
2. **Non-verbal communication** – sending messages without words, using facial expressions, gestures, posture, eye contact, body language and tone of voice.
3. **Visual communication** – conveying information through pictures, signs, symbols, charts, graphs, maps, posters and videos.
4. **Written communication** – communication through text such as notes, reports, letters, emails and books. It provides a permanent record.

The method is chosen according to the situation, the audience and the purpose of the message.

Q3. What is verbal communication? Give two examples.

Answer:

Verbal communication is the exchange of information using words. It can be **oral** (spoken) or **written**. It is the most common form of communication and is used to talk, discuss, explain and give instructions.

Examples:

1. A teacher explaining a lesson to students in the classroom (oral).
2. Writing an email or a letter to a friend or an office (written).

Q4. What is non-verbal communication? Give two examples.

Answer:

Non-verbal communication is the sending and receiving of messages without using spoken or written words. It includes facial expressions, gestures, eye contact, posture, body movements, touch and even tone of voice. It often supports and sometimes changes the meaning of the spoken words.

Examples:

1. Smiling to show happiness or friendliness.
2. Nodding the head to show agreement, or shaking the head to say "no".

Q5. What is visual communication?

Answer:

Visual communication is a method of conveying information through visual elements instead of only words. It includes pictures, symbols, signs, diagrams, charts, graphs, posters, maps and videos.

It is very useful because visual information can be understood quickly, easily and even by people who speak different languages. **Examples:** traffic signs, a "No Smoking" symbol, and charts in a textbook.

Q6. What is a communication cycle?

Answer:

A **communication cycle** is the complete process through which a message is sent by a sender and received and understood by a receiver, followed by feedback. Its main elements are:

1. **Sender** – the person who wants to share the message.
2. **Encoding** – converting the idea into words, gestures or symbols.
3. **Message** – the actual information being sent.
4. **Channel** – the medium used, e.g., speech, phone, letter, email.
5. **Receiver** – the person who gets the message.
6. **Decoding** – the receiver interprets and understands the message.
7. **Feedback** – the response of the receiver, which tells the sender whether the message was understood.

The cycle continues until the message is clearly understood by both sides.

Q7. What is feedback? Why is it important in communication?

Answer:

Feedback is the response given by the receiver after receiving and understanding a message. It can be verbal (saying "Yes, I understood"), written (a reply to an email) or non-verbal (nodding, smiling).

Importance of feedback:

- It confirms whether the message has been received and understood correctly.
- It helps the sender identify misunderstandings and clarify the message.
- It completes the communication cycle and makes communication two-way.
- It helps improve future communication and builds better relationships.

Q8. What are the common barriers to effective communication?

Answer:

Barriers are the obstacles that stop a message from being understood correctly. Common barriers are:

1. **Physical barriers** – noise, distance, poor telephone/internet connection.
2. **Language barriers** – different languages, difficult words, jargon or unclear speech.
3. **Emotional barriers** – anger, fear, stress or lack of trust.
4. **Cultural barriers** – different customs, beliefs and gestures in different cultures.
5. **Attitude/attention barriers** – poor listening, lack of interest, prejudice.
6. **Organisational barriers** – unclear message, wrong channel, too many levels of communication.

Ways to overcome: use simple language, listen actively, choose a suitable channel, give and take feedback, and communicate in a quiet environment.

Q9. What are the 7 Cs of effective communication?

Answer:

The 7 Cs are seven principles that make communication effective:

1. **Clear** – the message should be easy to understand with a clear purpose.
2. **Concise** – keep it short; avoid unnecessary words and repetition.
3. **Concrete** – be specific, give facts and figures, not vague statements.
4. **Correct** – information, grammar and spelling should be accurate.
5. **Coherent** – the message should be logical and well connected.
6. **Complete** – include all the necessary information the receiver needs.
7. **Courteous** – be polite, respectful and friendly.

Q10. How does body language affect communication?

Answer:

Body language is a part of non-verbal communication that includes facial expressions, gestures, posture, eye contact and body movements. It affects communication in the following ways:

- **Shows emotions and attitude** – a smile shows friendliness, crossed arms may show disagreement.
- **Builds confidence and trust** – good eye contact and upright posture make a speaker appear confident and honest.
- **Supports spoken words** – hand gestures help explain and emphasise a point.
- **Can create misunderstanding** – if body language does not match the words (e.g., saying "I am fine" with a sad face), the listener gets confused.
- **Shows interest** – leaning forward and nodding show that the listener is paying attention.

Hence, positive body language makes communication clear and effective, while negative body language weakens the message.

Unit 2: Self-Management Skills-II

Q11. What is stress?

Answer:

Stress is the physical, mental or emotional reaction of our body to pressure, challenges or demanding situations. It occurs when a person feels that the demands of a situation are more than what he/she can handle.

A small amount of stress can motivate us to finish work, but too much stress affects concentration, sleep, health and daily activities. **Example:** feeling nervous and unable to sleep before an examination.

Q12. What are the common causes of stress among students?

Answer:

Common causes of stress among students are:

1. **Examinations and results** – fear of failure or low marks.
2. **Academic pressure** – too much homework, assignments and projects.
3. **High expectations** – pressure from parents, teachers or self.
4. **Competition and comparison** with classmates.
5. **Poor time management** – doing everything at the last minute.
6. **Lack of sleep, unhealthy food and no physical activity.**
7. **Personal problems** – family issues, friendship problems, bullying or social media pressure.

Q13. Write any three techniques for managing stress.

Answer:

1. **Time management** – making a timetable, setting priorities and completing tasks on time avoids last-minute pressure.
2. **Physical exercise** – walking, yoga, sports or any activity releases tension, improves mood and keeps the body healthy.
3. **Meditation / deep breathing** – calms the mind, improves concentration and reduces anxiety.

Other techniques: adequate sleep, healthy diet, hobbies, positive thinking and talking to a trusted friend, parent or teacher.

Q14. How can meditation and physical exercise help in managing stress?

Answer:

Meditation	Physical Exercise
Calms the mind and relaxes the body.	Releases "feel-good" hormones (endorphins) that improve mood.
Improves focus, attention and memory.	Reduces muscle tension and tiredness.
Helps control negative thoughts and anxiety.	Improves sleep, energy and overall health.
Only 10–15 minutes of daily practice gives benefits.	Examples: walking, running, yoga, cycling, playing games.

Together with proper sleep and time management, both help a student stay calm, healthy and stress-free.

Q15. What is self-awareness?

Answer:

Self-awareness is the ability to know and understand one's own thoughts, emotions, strengths, weaknesses, interests, values and behaviour. It also means understanding how our actions affect other people.

A self-aware student can identify areas to improve, make better decisions and set realistic goals.

Example: a student who knows that he is weak in Maths gives extra time to practise it.

Q16. What is self-motivation?

Answer:

Self-motivation is the inner drive that encourages a person to start a task and keep working towards a goal without needing pressure or encouragement from others.

A self-motivated person stays focused even when work is difficult or boring. **Example:** a student who studies regularly on his own, without being told by parents.

Q17. What is self-regulation?

Answer:

Self-regulation is the ability to control and manage our thoughts, emotions, behaviour and habits according to the situation, so that we can achieve our goals.

It helps a person avoid impulsive actions, stay disciplined and remain calm under pressure. **Example:** a student controlling the urge to use a mobile phone while studying.

Q18. Why is self-motivation important for achieving goals?

Answer:

Self-motivation is important for achieving goals because:

- Goals need **continuous effort and patience**; self-motivation keeps a person going.
- It helps a person **overcome difficulties and failures** without giving up.
- It makes a person **focused and disciplined**, so tasks are completed on time.
- It **builds confidence** and a sense of responsibility.
- A self-motivated person **does not depend on others** for encouragement, so progress is regular.

Q19. What is the importance of working independently?

Answer:

Working independently means completing tasks and taking decisions on one's own without depending completely on others. Its importance:

- Develops **self-confidence** and **responsibility**.
- Improves **decision-making and problem-solving** skills.
- Helps us understand our **own strengths and weaknesses**.
- Builds **discipline and time management**.
- Prepares a person for **future studies and jobs** where independent work is expected.

Unit 3: ICT Skills-II

Q20. What is an operating system? Give two examples.

Answer:

An **Operating System (OS)** is the main system software of a computer that manages all the hardware and software resources and provides an interface between the user and the computer. Without an OS, a computer cannot work.

Main functions: managing memory, files, input/output devices and processes, running applications, and providing security (user login, passwords).

Examples: Microsoft Windows, Linux (Ubuntu), macOS, Android (for mobiles).

Q21. What is a file and what is a folder?

Answer:

File: A file is a collection of related data or information stored on a computer with a name and an extension. Examples: *notes.docx* (document), *photo.jpg* (image), *song.mp3* (audio).

Folder: A folder (also called a directory) is a container used to store and organise files and other sub-folders. Example: a folder named "Class 10" containing all subject files.

Q22. Differentiate between a file and a folder.

Answer:

Basis	File	Folder
Meaning	A collection of data or information stored on a computer.	A container used to store and organise files and sub-folders.
Content	Contains actual data – text, image, audio, video, etc.	Contains files and other folders, not data directly.
Extension	Has an extension, e.g., .docx, .jpg, .mp3.	Has no extension.
Size	Has its own size depending on data.	Size is the total size of files inside it.
Created by	An application (MS Word, Paint, etc.).	Operating system (New > Folder).
Example	science_notes.docx, image.png	"Class 10 Science", "My Photos"

Q23. What is file organization?

Answer:

File organisation means arranging files and folders on a computer in a systematic and logical way so that they can be found, used and managed easily.

Good file organisation uses **meaningful file names**, **separate folders** for different subjects/topics and **sub-folders** where needed. **Benefits:** saves time, avoids confusion, prevents data loss and makes backup easy.

Q24. What are the basic operations performed on files and folders?

Answer:

The basic operations that can be performed on files and folders are:

1. **Create** – making a new file or folder (Right-click > New).
2. **Open** – viewing the contents by double-clicking.
3. **Rename** – changing the name (F2 key or Right-click > Rename).
4. **Copy and Paste** – making a duplicate at another location (Ctrl+C, Ctrl+V).
5. **Cut and Paste / Move** – shifting to another location (Ctrl+X, Ctrl+V).
6. **Delete** – removing a file/folder; it goes to the Recycle Bin.
7. **Search** – finding a file/folder by its name.
8. **Sort / View** – arranging files by name, date, size or type.

Q25. Why is computer maintenance important?

Answer:

Computer maintenance means taking regular care of the computer's hardware and software so that it works properly. It is important because:

- It keeps the computer **fast and efficient** by removing unwanted files and programs.
- It **protects data** from viruses and malware through updated antivirus software.
- It **increases the life** of the computer and its parts (keeping it clean, dust-free and cool).
- It prevents **data loss** through regular backups.
- It keeps software **updated and secure**, fixing bugs and security problems.
- It **saves money** by avoiding expensive repairs and replacement.

PART B – SUBJECT SPECIFIC SKILLS (ARTIFICIAL INTELLIGENCE)

Unit 1: Revisiting AI Project Cycle & Ethical Frameworks for AI

Q26. What is the AI Project Cycle?

Answer:

The **AI Project Cycle** is a step-by-step framework that guides the development of an AI project, from understanding a real-world problem to building and testing an AI solution. It gives a proper structure so that the project is completed in an organised way.

It has five main stages: **Problem Scoping** → **Data Acquisition** → **Data Exploration** → **Modelling** → **Evaluation**. (Deployment is sometimes added as the last stage.)

Q27. Name the major stages of the AI Project Cycle.

Answer:

1. **Problem Scoping** – understanding and defining the problem using the 4Ws canvas (Who, What, Where, Why).
2. **Data Acquisition** – collecting reliable and relevant data from sources like surveys, sensors, websites, APIs, observations.
3. **Data Exploration** – cleaning the data and understanding it using visual tools like graphs and charts.
4. **Modelling** – selecting and training an AI model (rule-based or learning-based) on the data.
5. **Evaluation** – testing the model on new data to check its accuracy and reliability.
6. **Deployment** – (optional stage) putting the model to actual use in the real world.

Q28. What is problem scoping?

Answer:

Problem scoping is the first stage of the AI Project Cycle in which we clearly identify and define the problem that the AI project will solve. It involves understanding the situation, the people affected, the goal and the limitations.

The **4Ws Problem Canvas** is used for problem scoping:

- **Who** – Who is facing the problem? (stakeholders)
- **What** – What is the exact problem? What is the evidence?
- **Where** – Where/in what situation does the problem occur?
- **Why** – Why is it important to solve? What benefit will it give?

Finally, a **Problem Statement** is written summarising all the four Ws.

Q29. Why is problem scoping important in an AI project?

Answer:

Problem scoping is important because:

- It gives a **clear goal and direction** to the whole project; an unclear problem leads to an unsuitable solution.
- It identifies the **stakeholders** (people affected) and their real needs.
- It helps decide **what data** will be needed and **which AI approach** to use.
- It sets the **limits (scope)** of the project so that time and resources are not wasted.
- It provides a **strong foundation** – all the later stages depend on it.

Q30. What are the three domains of AI? Give one example of each.

Answer:

Domain	What it deals with	Example
Data Science / Statistical Data	Works with numerical and text data (tables, records) to find patterns and make predictions.	Predicting sales of a shop, price prediction, recommendation systems.
Computer Vision (CV)	Works with images and videos; helps machines "see" and understand visual data.	Face recognition / Face lock in phones, self-driving cars.
Natural Language Processing (NLP)	Works with human language (text and speech); helps machines understand and respond to language.	Chatbots, Google Translate, voice assistants like Alexa and Siri.

Q31. What is an ethical framework in AI?

Answer:

An **ethical framework in AI** is a set of principles, values and guidelines that help developers and users to design, build and use AI systems in a responsible, fair and safe manner.

It considers questions like: Is the AI fair to everyone? Does it protect privacy? Who is responsible if it makes a mistake? Main principles include:

- **Fairness** – no discrimination or bias against any group.
- **Transparency** – how the AI makes decisions should be clear.
- **Privacy** – personal data should be protected.
- **Accountability** – someone should be responsible for the AI's actions.
- **Safety and human well-being** – AI should not harm people.

Q32. Why are ethics important in Artificial Intelligence?

Answer:

Ethics are important in AI because AI systems affect people's lives, decisions, jobs and privacy. The main reasons are:

- **To prevent bias and discrimination** – AI trained on biased data can be unfair to some groups.
- **To protect privacy** – AI uses large amounts of personal data which must be kept safe.
- **To ensure transparency** – people should know how and why AI took a decision.
- **To fix responsibility** – e.g., who is responsible if a self-driving car causes an accident.
- **To prevent misuse** – like fake videos (deepfakes), cheating or surveillance.
- **To build trust** – people will accept AI only if it is safe and responsible.

Q33. What is AI bias? Give an example.

Answer:

AI bias occurs when an AI system gives unfair or one-sided results towards a particular group of people. It usually happens because the **training data is biased or incomplete**, or due to the assumptions of the developers. AI learns from data – if the data is biased, the AI will also be biased.

Examples:

- A hiring AI trained mostly on data of male employees started rejecting female candidates.
- A face recognition system trained mostly on fair-skinned faces failed to recognise dark-skinned faces properly.
- Virtual assistants usually having female voices, showing gender stereotypes.

Prevention: use diverse and balanced data, and test the model on different groups of people.

Unit 2: Advanced Concepts of Modelling in AI

Q34. What is Artificial Intelligence?

Answer:

Artificial Intelligence (AI) is a branch of computer science that creates machines and software that can think, learn and perform tasks that normally require human intelligence, such as understanding language, recognising images, reasoning, learning from experience and making decisions.

An AI machine collects data, finds patterns in it, learns from it and improves its performance over time.

Examples: Google Maps, chatbots, face lock, recommendation on YouTube/Netflix, self-driving cars.

Q35. Differentiate between AI, Machine Learning and Deep Learning.

Answer:

Basis	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Meaning	Broad field of making machines that can perform intelligent tasks like humans.	A subset of AI in which machines learn patterns from data without being explicitly programmed.	A subset of ML that uses artificial neural networks with many layers to learn complex patterns.
Scope	Widest – includes ML and DL.	Narrower – part of AI.	Narrowest – part of ML.
Data required	May work with rules or data.	Needs a moderate amount of data.	Needs very large amounts of data.
Human role	Rules may be written by humans.	Humans select features; machine learns.	Machine itself extracts features from data.
Hardware	Normal computers.	Normal computers.	Needs powerful computers (GPUs).
Example	Chess-playing program, chatbot.	Spam email filter, price prediction.	Face recognition, self-driving cars, voice assistants.

Relationship: Deep Learning $\subset$ Machine Learning $\subset$ Artificial Intelligence.

Q36. What is a rule-based approach in AI?

Answer:

In a **rule-based approach**, the machine follows a set of rules that are defined by the programmer. The rules are usually written in the form of **IF–THEN** statements, and the machine cannot go beyond these rules or learn anything new by itself.

Example: IF temperature > 40°C THEN display "Heat-wave warning". A traffic light system or a simple quiz-marking program are also rule-based.

Limitation: It is *static* – if a new situation arises that is not covered by the rules, the system fails and rules must be added manually.

Q37. What is a learning-based approach?

Answer:

In a **learning-based approach**, the machine is **trained with data** and learns the patterns and relationships by itself, instead of following rules written by a human. Once trained, it can make predictions or decisions on new, unseen data and can improve with more data.

Machine Learning and Deep Learning are learning-based approaches. It is *dynamic* – it can adapt to new situations.

Example: A model trained on thousands of photos of apples and bananas learns to identify a new fruit photo on its own. Types: Supervised, Unsupervised and Reinforcement learning.

Q38. What is supervised learning? Give an example.

Answer:

Supervised learning is a type of machine learning in which the model is trained on **labelled data**, i.e., each input in the training data has its correct output (label) attached. The model learns the relationship between input and output and then predicts the output for new data. It is like learning under the guidance of a teacher.

Two types: Classification (output is a category) and Regression (output is a number).

Example: Training a model with thousands of emails already marked as "spam" or "not spam"; later the model classifies a new email as spam or not spam. Another example: predicting house prices from area and location.

Q39. What is unsupervised learning? Give an example.

Answer:

Unsupervised learning is a type of machine learning in which the model is trained on **unlabelled data** – no correct answers are given. The model itself finds hidden patterns, structures or groups in the data. There is no "teacher" to guide it.

Main techniques: Clustering (grouping similar data) and Dimensionality Reduction.

Example: An online store gives the model its customers' purchase data without any labels; the model automatically groups customers with similar buying habits so the store can plan offers for each group.

Q40. Differentiate between classification and regression.**Answer:**

Basis	Classification	Regression
Meaning	Predicts a category / class to which the data belongs.	Predicts a continuous numerical value .
Type of output	Discrete (Yes/No, Spam/Not Spam, Cat/Dog).	Continuous (price, temperature, marks, salary).
Type of learning	Supervised learning.	Supervised learning.
Nature of data	Labelled with class names.	Labelled with numeric values.
Evaluation	Accuracy, precision, recall, F1 score.	Mean squared error, mean absolute error.
Example	Predicting whether a patient has a disease or not.	Predicting tomorrow's temperature or house price.

Q41. What is clustering? Give a suitable example.**Answer:**

Clustering is an **unsupervised learning** technique in which the model divides unlabelled data into groups called **clusters**, such that data items in the same cluster are similar to each other and different from those in other clusters. The model itself discovers these groups without any pre-defined categories.

Example: An online shopping website clusters its customers on the basis of their purchasing patterns – e.g., "frequent buyers", "festival buyers" and "discount seekers" – so that suitable offers can be sent to each group. Another example: grouping news articles by topic.

Q42. What is an Artificial Neural Network (ANN)?**Answer:**

An **Artificial Neural Network** is a machine-learning model inspired by the structure and working of the **human brain**. It is made up of many interconnected units called **neurons (nodes)**, arranged in layers. It is the base of Deep Learning.

Layers of a neural network:

1. **Input layer** – receives the raw data (e.g., pixels of an image).
2. **Hidden layer(s)** – process the data; each layer extracts features and passes the result forward. A network with many hidden layers is called a *deep* neural network.
3. **Output layer** – gives the final result or prediction.

During training the network adjusts the **weights** of connections to reduce errors. **Uses:** image recognition, speech recognition, language translation, prediction.

Q43. What is a Convolutional Neural Network (CNN)? Where is it commonly used?

Answer:

A **Convolutional Neural Network (CNN)** is a special type of deep neural network designed to process **grid-like data such as images**. It uses a mathematical operation called **convolution** to automatically detect features like edges, shapes, colours and patterns from an image.

Main layers of a CNN:

1. **Convolution layer** – applies filters (kernels) to extract features from the image.
2. **ReLU layer** – removes negative values and adds non-linearity.
3. **Pooling layer** – reduces the size of the image while keeping important features.
4. **Fully connected layer** – combines all features and gives the final classification.

Commonly used in: image classification, object detection, face recognition, medical image analysis (X-ray, MRI), self-driving cars and handwriting recognition – i.e., in the domain of Computer Vision.

Unit 3: Evaluating Models

Q44. What is model evaluation in AI?

Answer:

Model evaluation is the process of checking how well a trained AI model performs and how reliable its predictions are. It is the last stage of the AI Project Cycle. The model is tested on **testing data** (data it has not seen during training) and its predictions are compared with the actual results.

Evaluation is done using **metrics** like Accuracy, Precision, Recall and F1 Score, usually with the help of a Confusion Matrix. It tells us whether the model is ready for use or needs improvement.

Q45. Why is it important to evaluate an AI model?

Answer:

Evaluating an AI model is important because:

- It tells us **how accurate and reliable** the model is before it is used in real life.
- A model may perform well on training data but poorly on new data (**overfitting**); evaluation detects this.
- It helps to **identify errors and weaknesses** so the model can be improved.
- It allows us to **compare different models** and select the best one.
- It checks whether the model is **fair** and works equally well for all groups.
- It builds **trust** – a tested model can be safely deployed.

Q46. What is Train-Test Split?

Answer:

Train-Test Split is a method of dividing the available dataset into two separate parts before building a model:

- **Training set** (usually 70–80%) – used to train the model, i.e., the model learns patterns from this data.
- **Testing set** (usually 20–30%) – kept separate and used after training to check how well the model performs on unseen data.

Why it is needed: If the same data is used for training and testing, the model may just "memorise" the answers and give a falsely high score. Testing on separate data gives a true and realistic measure of the model's performance.

Q47. What is accuracy? Write its formula.

Answer:

Accuracy is the percentage of correct predictions made by a model out of the total predictions. It considers both correctly predicted positive cases (TP) and correctly predicted negative cases (TN).

Formula:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \times 100 \%$$

or **Accuracy = (Correct predictions / Total predictions) × 100 %**

Example: If a model makes 100 predictions and 90 are correct, $\text{Accuracy} = 90 / 100 \times 100 = 90\%$.

Note: Accuracy is a good measure only when the classes are balanced.

Q48. What is a confusion matrix? Explain TP, TN, FP and FN.

Answer:

A **confusion matrix** is a table that is used to evaluate the performance of a classification model by comparing the **actual values** (reality) with the **predicted values** of the model. It has four parts:

	Predicted: Yes (Positive)	Predicted: No (Negative)
Actual: Yes	True Positive (TP)	False Negative (FN)
Actual: No	False Positive (FP)	True Negative (TN)

- **True Positive (TP)** – Actual is Yes and model predicted Yes. (*Correct*) e.g., it rained and the model predicted rain.
- **True Negative (TN)** – Actual is No and model predicted No. (*Correct*) e.g., no rain and the model predicted no rain.
- **False Positive (FP)** – Actual is No but model predicted Yes. (*Wrong – Type I error*) e.g., no rain but the model predicted rain.
- **False Negative (FN)** – Actual is Yes but model predicted No. (*Wrong – Type II error*) e.g., it rained but the model predicted no rain.

From these four values, Accuracy, Precision, Recall and F1 Score are calculated.

Q49. What are precision, recall and F1 Score?

Answer:

These are evaluation metrics used to measure the performance of a classification model:

1. **Precision** – Out of all the cases the model predicted as positive, how many were actually positive. It shows how *reliable* the positive predictions are.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

2. **Recall** – Out of all the actual positive cases, how many the model correctly identified. It shows how *complete* the model is in finding positives.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

3. **F1 Score** – The harmonic mean of Precision and Recall. It balances both and is useful when we need a single score.

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The F1 score ranges from 0 to 1; a value close to 1 means the model is good. A model is considered good only when both precision and recall are high.

Q50. Differentiate between precision and recall. Also explain why accuracy alone may not always be sufficient for evaluating an AI model.

Answer:

(a) Difference between Precision and Recall:

Basis	Precision	Recall
Meaning	Out of all predicted positives, how many are actually positive.	Out of all actual positives, how many were correctly predicted.
Formula	$TP / (TP + FP)$	$TP / (TP + FN)$
Focus	Reduces False Positives (false alarms).	Reduces False Negatives (missed cases).
Question it answers	"How correct are the positive predictions?"	"How many real positives did we catch?"
Important when	Cost of a false alarm is high – e.g., spam filter (a genuine mail must not go to spam).	Cost of missing a case is high – e.g., disease detection (no patient should be missed).

(b) Why accuracy alone is not sufficient:

- Accuracy is **misleading when the data is imbalanced** (one class is much larger than the other).
- **Example:** Suppose 1000 people are tested for a rare disease and only 10 actually have it. A model that predicts "No disease" for everyone will be 99% accurate but will miss all 10 patients – its recall is 0. Such a model is useless even though accuracy is high.
- Accuracy **does not tell which type of error** (FP or FN) the model is making, and in real life different errors have different costs.
- Therefore, Precision, Recall and F1 Score must also be checked, along with the confusion matrix, to judge a model properly.